Treatment Cost Estimates for Bd Interventions in *Litoria aurea*

Overview

This report estimates the economic costs of the two treatment interventions in the ODE model: frog saunas (T1 compartment, control variable ζ) and antifungal itraconazole baths (T2 compartment, control variable ξ). All costs are presented in both AUD and USD (using approximate exchange rate of 1 AUD ≈ 0.65 USD as of early 2026). These estimates are critical for any cost-effectiveness or optimal control analysis of the model.

---

1. Frog Sauna (Hotspot Shelter) — T1

1.1 Construction Cost

Anthony Waddle, the lead author of the Nature study, has stated directly that the materials cost approximately **AUD $70 (~USD $45)** per unit. The construction guide (Waddle & Shine, hosted by The Conversation) specifies the exact materials:

Component	Specification	Approx. AUD cost	Source
Drop-over greenhouse	Greenlife 900×900×1020mm PE cover	$35–45	Bunnings / Greenlife Australia
Masonry bricks (×10–14)	CSR PGH 230×110×76mm 10-hole clay bricks	$15–20 (~$1.50 each)	Bunnings
Black spray paint (×1–2 cans)	Rust-Oleum 340g flat black 2× Ultra Cover	$8–12	Bunnings
Cable ties (100-pack)	Crescent 300×4.8mm	$3–5	Bunnings
Total materials		**~AUD $65–80**	
Central estimate		**AUD $70 (USD $45)**	Waddle, CBC interview 2024

Labor: Assembly is extremely simple — paint bricks, assemble greenhouse frame, stack bricks, drop cover over. A single person can build one in approximately **30–60 minutes** with no specialized tools. At a conservation technician wage of ~AUD $35/hr, labor adds roughly **AUD $20–35** per unit.

Total installed cost per sauna: AUD $90–115 (USD $60–75)

For a large-scale deployment (e.g., 50 units as Waddle has installed at Sydney Olympic Park), bulk purchasing of bricks and greenhouses would likely reduce per-unit costs by 10–20%.

1.2 Operational Lifespan and Maintenance

The Waddle & Shine construction guide notes that shelters work best in autumn/winter and could be "seasonally repurposed as a garden" in summer. Key durability considerations:

PE (polyethylene) greenhouse cover: This is the weakest link. Standard PE greenhouse covers degrade under UV exposure and are rated for approximately **2–4 seasons** (i.e., 2–4 years) before becoming brittle and requiring replacement. In Sydney's climate with moderate UV, expect **~3 years** before cover replacement. Replacement PE covers for Greenlife drop-over greenhouses cost approximately **AUD $15–25**.

Steel frame: The galvanized steel frame of the Greenlife greenhouse is rated for outdoor use and should last **5–10+ years** with minimal corrosion.

Bricks: Masonry clay bricks are effectively permanent in outdoor conditions. They will not degrade meaningfully within **20+ years**.

Paint: Black paint on bricks may fade or peel after **2–3 years** of UV and weather exposure, requiring touch-up or repainting (~AUD $8 per can, one can does ~10 bricks).

Wind damage: Waddle's guide emphasizes securing frames against high winds with cable ties to extra bricks or tent pegs. Wind damage to the PE cover is probably the most common maintenance issue.

Maintenance item	Frequency	Cost per event (AUD)
PE cover replacement	Every 2–4 years	$15–25
Paint touch-up	Every 2–3 years	$8–12
Cable tie / securing repair	As needed (~annually)	$3–5
Annual maintenance cost		**~AUD $10–15/year**

Effective lifespan without any maintenance: Approximately **2–3 years** before the PE cover degrades sufficiently to lose the greenhouse effect. After cover failure, the painted bricks alone still provide some thermal benefit (dark surfaces absorb sunlight), but without the greenhouse trapping, internal temperatures will not reliably reach the 29–30°C needed for Bd clearance.

Effective lifespan with basic maintenance (cover replacement): 5–10+ years (limited by frame corrosion).

1.3 Cost Per Frog Treated (T1)

Each frog sauna can serve multiple frogs simultaneously — the 10-brick stack has ~100 holes, and Waddle observed frogs using shelters at **4× the expected rate**. At Sydney Olympic Park, 50 shelters have been deployed for a population estimated in the low hundreds. A rough capacity estimate:

- Each shelter can accommodate **5–15 frogs** at any time (conservative, given 100 holes per 10-brick stack and observed occupancy patterns)
- Over one winter season (~May–September, ~150 days), with frogs cycling through 6-day clearance periods, each shelter could treat roughly **25–50 unique frogs per season** (assuming each frog uses the sauna for ~6 days before clearing infection, with some shelters more popular than others)

Amortized cost per frog-treatment via sauna:

Scenario	Frogs treated/season	Total cost over 3 years	Cost per treatment (AUD)
Low use (10 frogs/season)	30 over 3 years	$105 (construction + maintenance)	**~$3.50**
Medium use (25 frogs/season)	75 over 3 years	$105	**~$1.40**
High use (50 frogs/season)	150 over 3 years	$105	**~$0.70**

The frog sauna is remarkably cost-effective — on the order of **AUD $1–4 per frog-treatment** depending on population density and shelter use rates.

---

2. Antifungal Itraconazole Bath — T2

2.1 Drug Cost

The standard Bd treatment protocol uses Sporanox (itraconazole) 10 mg/mL oral solution, diluted to create a bath.

Drug pricing:

Source	Price for 150 mL bottle	Per mL
The Animal Pharmacy (Australia)	AUD $254.90	AUD $1.70/mL
US generic (Cost Plus Drugs / GoodRx)	~USD $50–80 (150 mL)	~USD $0.33–0.53/mL
US brand Sporanox	~USD $200–400 (150 mL)	~USD $1.33–2.67/mL

Drug consumption per frog-treatment:

The optimized Brannelly et al. (2012) protocol: 0.0025% itraconazole, 5-minute bath daily for 6 days.

- 0.0025% = 25 mg/L = 25 μg/mL
- From 10 mg/mL stock: dilute 1 mL stock into 399 mL amphibian Ringer's solution (or ~2.5 mL per liter)
- Bath volume per frog: approximately **20–50 mL** of dilute solution (small container for one frog)
- Stock solution consumed per frog per day: **0.05–0.125 mL** (at 2.5 mL stock per liter, using 20–50 mL bath)
- Over 6-day treatment: **0.3–0.75 mL of stock solution per frog**

Item	Quantity per frog (6-day treatment)	Cost (AUD)
Itraconazole stock (0.5 mL avg)	0.3–0.75 mL from 150 mL bottle	**$0.50–1.30**
Amphibian Ringer's solution	~200 mL total (reusable components)	**$0.50–1.00**
Disposable containers/swabs	6 per treatment	**$1.00–2.00**
Drug & materials subtotal		**AUD $2.00–4.30**

One 150 mL bottle of Sporanox can treat approximately **200–500 individual frogs** at the optimized low-dose protocol. This makes the per-frog drug cost remarkably low.

2.2 Labor and Field Costs (The Dominant Expense)

The antifungal bath treatment is labor-intensive because every frog must be individually captured, held in captivity for 6 days of daily baths, and then released. This is where the cost of ξ (capture rate) becomes economically significant.

Field capture costs:

Activity	Time estimate	Personnel	Cost (AUD)
Night survey to locate/capture frogs	3–4 hours per session	2 technicians @ $35/hr	$210–280 per session
Frogs captured per session	5–15 (depending on population density)		
Transport to treatment facility	1 hour round trip	1 vehicle + 1 person	$50–80
Per-frog capture cost			**AUD $20–70**

Treatment holding costs (6-day protocol):

Item	Per frog	Cost (AUD)
Individual housing (plastic container, substrate, water changes)	6 days	$5–10
Daily itraconazole bath administration (5 min/frog + prep)	~15 min/frog/day × 6 days = 1.5 hrs total	$52 (technician time)
Post-treatment qPCR swab + testing	1–2 swabs	$15–30
Feeding (crickets, 6 days)	Small quantity	$2–5
Treatment holding subtotal		**AUD $75–95**

2.3 Total Distributed Cost Per Frog — Antifungal Bath

Cost component	Range (AUD)	Central estimate (AUD)
Drug and materials	$2–4	$3
Field capture	$20–70	$40
Treatment holding (6 days)	$75–95	$85
Release and monitoring	$10–20	$15
Total per frog	**$107–189**	**~AUD $143 (USD $93)**

In a well-organized field program treating many frogs simultaneously, per-frog costs could drop to the lower end through economies of scale (batch capture, shared housing facilities, etc.). In a captive colony setting where frogs are already held, costs drop dramatically to essentially just drug + technician bath time ≈ **AUD $55–60 per frog**.

---

3. Comparative Cost-Effectiveness Summary

Parameter	Frog Sauna (T1)	Antifungal Bath (T2)
Capital cost per unit	AUD $70–115	N/A (facility-based)
Cost per frog-treatment	AUD $1–4	AUD $107–189 (field); $55–60 (captive)
Treatment duration	~6 days (self-administered)	6 days (daily human intervention)
Scalability	Excellent (passive, no labor per frog)	Poor (labor-intensive, one frog at a time)
Acquired immunity post-treatment	Yes — 23× survival advantage (Waddle 2024)	Partial — lower Bd loads persist, ~4× survival advantage (Cook 2022)
Side effects	None documented	22% weight loss (Hardy 2015); potential growth suppression
Ongoing maintenance	~AUD $10–15/year	Requires trained personnel for every treatment
Geographic applicability	Requires sunny climates reaching ⩾28°C	Universal (any climate, any setting)
Species applicability	Only heat-tolerant, basking species	Broad (most amphibians, with species-specific dosing)

Key insight for optimal control modeling

The frog sauna is approximately **30–130× cheaper per frog-treatment** than the antifungal bath in field settings, and it additionally confers stronger acquired immunity. However, the antifungal bath is universally applicable regardless of climate and species thermal tolerance. In the ODE model, this means:

- **Increasing ζ (frog sauna density) is far more cost-effective per unit of disease reduction** than increasing ξ (antifungal treatment rate), as long as the climate supports sufficient solar heating.

- The **optimal control strategy** likely involves deploying saunas as the primary intervention (maximizing ζ) and reserving antifungal baths (ξ) for targeted use on high-value individuals or populations where saunas are insufficient.
- In a cost-minimization framework, the cost function might look something like: $\mathbf{C(\zeta, \xi) = \$100 \cdot N_saunas + \$143 \cdot \xi \cdot I \cdot \Delta t}$ where N_saunas is the number of shelters deployed (each increasing ζ by some increment) and the second term is the per-frog treatment cost times the flow into T2.

---

4. Sources

1. Waddle AW & Shine R. "How to build a frog sauna." Construction guide, The Conversation / Macquarie University. PDF: cdn.theconversation.com
2. Waddle AW, et al. (2024) Hotspot shelters stimulate frog resistance to chytridiomycosis. *Nature* 631:344–349
3. CBC News (2024) "This DIY brick sauna is helping frogs fight disease." Interview with A. Waddle: materials cost ~AUD $70
4. Science/AAAS (2024) "Frog 'saunas' could help endangered species beat a deadly fungus." Reports 50 shelters deployed at Sydney Olympic Park
5. The Wildlife Society (2024) "Wild Cam: Frog saunas may ward off chytrid disease." Notes brick manufacturers now marketing bricks as frog saunas
6. Brannelly LA, Richards-Zawacki CL, Pessier AP (2012) Clinical trials with itraconazole as a treatment for chytrid fungal infections in amphibians. *Dis. Aquat. Org.* 101:95–104
7. Pessier AP & Mendelson JR (2010) A manual for control of infectious diseases in amphibian survival assurance colonies and reintroduction programs. IUCN/SSC CBSG. Chapter 8: Disease Treatment & Control
8. The Animal Pharmacy Australia — Sporanox Oral Solution 150mL pricing: AUD $254.90
9. Cook et al. (2022) In situ treatment of juvenile frogs for disease can reverse population declines. *Conserv. Sci. Pract.* 4:e12762
10. Hardy et al. (2015) Itraconazole treatment reduces Bd prevalence and increases overwinter field survival in juvenile Cascades frogs. *Dis. Aquat. Org.* 112:243–250
11. Greenlife Australia — Drop Over Greenhouse product page (greenlife.com.au)